

S.T. Yau High School Science Award (Asia) 2022

Research Report

Research on the Unsinkable Disk

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Research on the Unsinkable Disk

Abstract

A metal disk with a hole at its centre sinks in a container filled with water. However, when a vertical water jet hits the centre of the disc, it may float on the water surface. The various forces the floating disk primarily experiences include Weight, Downthrust of the water jet hitting the disk and an Upthrust force by water due to hydraulic jump, surface waves and the air entrained. The floating status of the disk is therefore determined by the relative magnitude of these forces. This research investigates the effects of the relative parameters on the floating status of an unsinkable disk, and suggests a possible theoretical model to explain such a phenomenon from both qualitative and quantitative approaches. Various experiments and computational simulations were conducted to verify the validity of the theoretical model. A comprehensive conclusion that the maximum mass of the disk the maximum mass of the disk the hydraulic jump can support is in positive relationship with the disk radius, but only in positive relationship with jet velocity, jet radius and hole radius initially. As the jet velocity, jet radius and hole radius increases pass a certain threshold, the maximum mass of the disk the hydraulic jump can support will decrease.

Keywords: Unsinkable Disk, Hydraulic Jump, Laminar flow, Turbulent flow, Calculating Fluid Dynamics

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Introduction

1.1 Background

This interesting phenomenon of unsinkable disk was first identified and published by A. Luzin in 1999 in *Quantum* Vol.10 No.1[1]. He explained qualitatively about the general reason for the disk to become unsinkable. However, the main contributor for the effect—the hydraulic jump, was noted and discussed much earlier by various physicists including Rayleigh(1921) and Watson(1964). In fact, hydraulic jump effects are more common in our daily life than many may perceive. It can just take place in our kitchen, when a smooth jet of water from the tap falls vertically on to the bottom of the empty sink. The water spreads out in a thin layer until a sudden increase of depth occurs, giving rise to the hydraulic jump. Despite being discovered a long time ago, the theory we use today to model hydraulic jump is still surprisingly complicated, mainly due to its turbulent nature adding difficulties for hydrodynamics calculations.

1.2 Literature Review

A. Luzin (1999) explained qualitatively about the general reason for the disk to become unsinkable. However, his work lacks quantitative analysis and did not specify how exactly does each parameter affect the floating status of the disk.

Belanger(1838) first applied fluid momentum equation on hydraulic jump, which is the main principle we are using the model the hydraulic jump

Rayleigh(1921) first noticed the formation of a thin layer of hydraulic jump. However, his discussion is limited to channel of constant width.

Watson(1964) derived the main principle for circular hydraulic jump on an infinitely large plane. However, I need to adapt his theory to apply to our case due to the hole in the middle of the disk.

Bechtoh(1967) first introduced the idea of Froude number into calculation of hydraulic jump. This greatly simplified our calculation.

Majid(2014) calculated the contact angle between surfaces of water and various metals, which helped us to obtain the approximate value of upward force due to surface tension on the disk.

Theory

2.1 Flow Analysis

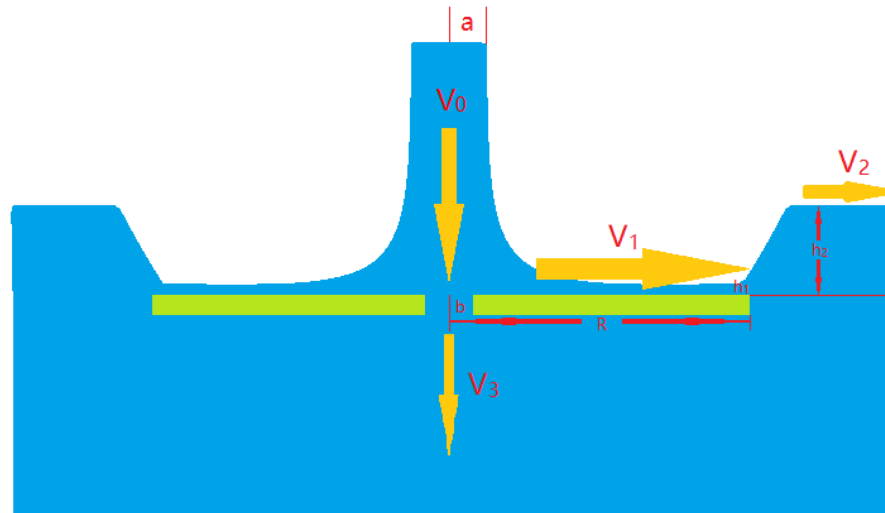


Figure 2.1: Water flow around the disk

Fig. 2.1 shows a vertical section of the phenomenon along the plane vertical to the disk's diameter. A vertical water jet with radius a strikes the centre of the disk with radius R and hole radius b . The velocity of the falling jet before striking the plane is v_0 . After hitting the plane, part of the water current will flow downwards through the hole with a velocity of v_3 . The rest of the water current will develop into a fast laminar flow of velocity v_1 along the axial direction. Upon reaching the end of the plate, the water current becomes turbulent and a hydraulic jump is developed. A slow laminar flow of velocity v_2 is observed after the hydraulic jump.

2.2 Surface Tension



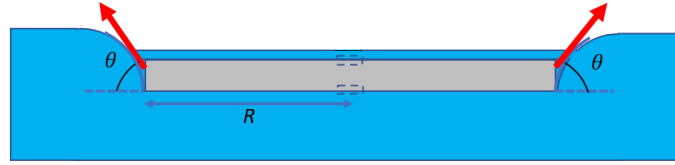


Figure 2.2: Surface tension effects on the disk

At the interface between a liquid and a gas, a solid or another immiscible liquids, forces develop in the liquid surface which cause the surface to behave as if it were a “skin” or “membrane” stretched over the fluid mass(Fig. 2.2). A tensile force may be considered to be acting in the plane of the surface along any line in the surface. The intensity of the force per unit length along any line in the surface is called the surface tension and is designated by the Greek symbol σ [4].

Hence by calculation, the upward force the disk experiences due to surface tension is

$$F_s = 2\pi R\sigma \sin\theta \quad (2.1)$$

Where θ is the angle of contact between the water surface and the disk's rim.

With reference to the International Tables for the Surface Tension of Water [1], At room temperature in Singapore which is 30°C , the surface tension of water, σ , has a value of 0.07120 N/m. And by the measurement done by Dr Majid [2], the value of the angle of contact θ , between aluminium and water surface is approximately 70.0°. We can therefore calculate that a for typical aluminium disk of radius 0.0625m we used in our experiments, the upward force it experiences is 0.026 N.

This force is too small to support the disk against the impact of the water jet. Additionally, when the disk is hit by the water jet, it is completely immersed in water so no net force due to surface tension acts on it. Hence, surface tension effect is not the main reason why the disk becomes unsinkable. However, it is worth mentioning that surface tension effect is an important characteristic of liquid, which must be considered by the CFD software we are going to use later to produce our computational model during its calculation.

2.3 Characteristic Flow Parameters

The Reynolds number is a measure of the ratio of the inertia force on an element of fluid to the viscous force on an element. When the Reynolds number is large, the flow is likely to be turbulent with small viscous effect relative to inertia effect. When the Reynolds number is small, the flow is likely to be laminar flow and the viscous effect cannot be ignored. It is generally defined as

$$Re = \frac{\rho u L}{\mu} = \frac{u L}{\nu} \quad (2.2)$$

where ρ is the density of the fluid in $kg\ m^{-3}$, u is the flow velocity in $m\ s^{-1}$, L is a characteristic linear dimension of the flow in m , μ is the dynamic viscosity of the fluid in $Pa\ s^{-1}$, and ν is the kinematic viscosity of the fluid in $m^2\ s^{-1}$.

However, in this case, Watson(1964) pointed out that the Reynolds number of the jet should be better calculated as

$$Re = \frac{Q}{\nu a} \quad (2.3)$$

Since the velocity of the jet we use in this experiment, v_0 , ranges from 1.50 to $6.50\ m\ s^{-1}$; diameter of the jet, a , ranges from $0.003\ m$ to $0.005\ m$, kinematic viscosity of water at $303.15K$ is $8.007 \times 10^{-7}\ m^2\ s^{-1}$ [7], the Reynolds number of the water jet is calculated to be from 17656 to 127516 . Such a high Reynolds number implies that turbulent flow will be dominant in our model, and it is reasonable for us to ignore viscous effects of water. As such, we shall apply the K - ϵ Turbulent Model in our CFD model.

Additionally, the Froude number is given by

$$Fr = \frac{u}{\sqrt{gl}} \quad (2.4)$$

where u is the local flow velocity in $m\ s^{-1}$ and l is characteristic linear dimension of the flow in m . It measures the ratio of the inertia force on an element of fluid to the weight of the element. Introducing the Froude number will be extremely helpful in our analysis for hydraulic jump.

The Mach number is given by

$$Ma = \frac{u}{a} \quad (2.5)$$

where u is the local flow velocity in $m\ s^{-1}$ and a is the local speed of sound. Hence the Mach number of flow in our experiments ranges between $0.0044\ Ma$ to $0.019\ Ma$, far below the theoretical threshold value of 0.3 . Therefore, it is reasonable to consider water as incompressible in our experiments.

2.4 Hydraulic Jump



Figure 2.2: Circular hydraulic Jump

The main lifting force experienced by the disk comes from a circular hydraulic jump. A hydraulic jump is a fluid shock wave created at the transition between laminar and turbulent flow [2]. The water initially flows in a smooth sheet with consistent current patterns along the radial direction on the disk surface. Upon reaching the edge, an abrupt change occurs. The water depth increases and flow becomes turbulent. A difference in the water level results in larger pressure being exerted on the lower surface of the disk, resulting in a lifting force on the disk. Rayleigh first described the problem in 1914 [3]. He calculated the change in fluid depth associated with the shockwave and introduced the concepts of continuity and fluid momentum conservation in his derivation.

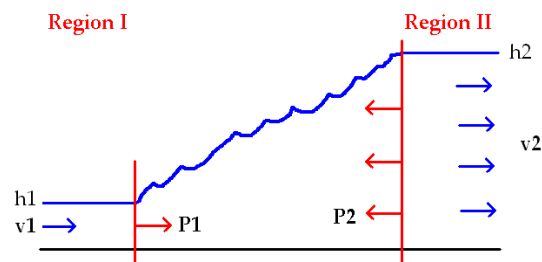


Figure 2.3: Vertical section of the flow pattern at the hydraulic jump

Continuity of fluid flow implies that the volume flow rate, $Q = hv$, must be equal before and after the hydraulic jump (Fig. 2.3):

$$Q = h_1 v_1 = h_2 v_2 \quad (2.1)$$

Belanger(1838) first applied fluid momentum equation on this phenomenon[3]. Across the hydraulic jump, the thrust of the pressure is equal to the rate at which momentum is destroyed.

The thrust of the pressure across the jump is given by

$$F_1 - F_2 = \frac{1}{2} \rho g (h_1^2 - h_2^2) \quad (2.2)$$

The rate of loss of momentum is given by

$$-\frac{dp}{dt} = \rho Q (v_2 - v_1) \quad (2.3)$$

Hence, we obtain

$$\frac{1}{2} \rho g (h_1^2 - h_2^2) = \rho Q (v_2 - v_1) \quad (2.4)$$

Defining the ratio of fluid depth before and after the hydraulic jump, $K = \frac{h_2}{h_1}$, and incorporating the fluid Froude number at the start of the hydraulic jump, $Fr = \frac{v_1}{\sqrt{gh_1}}$, we can rearrange this equation into

$$K^2 + K - 2Fr^2 = 0 \quad (2.5)$$

Solving this equation yields:

$$K = \frac{h_2}{h_1} = \frac{1}{2} \left(-1 + \sqrt{1 + 8Fr^2} \right) \quad (2.6)$$

Hence, the height of water after the hydraulic jump is

$$h_2 = \frac{1}{2} h_1 \left(-1 + \sqrt{1 + \frac{8v_1^2}{gh_1}} \right) \quad (2.7)$$

2.5 K - ε Turbulent Model

The K - ε Turbulent Model is governed by the Navier-Stokes Equation, which is a system of partial differential equations that describe the motion of a fluid in space. It has a general form of

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (2.8)$$

$$\rho \frac{\partial \mathbf{u}}{\partial t} + \rho (\mathbf{u} \cdot \nabla) \mathbf{u} = \nabla \cdot [-p \mathbf{I} + \boldsymbol{\tau}] + \mathbf{F} \quad (2.9)$$

$$\rho C_p \left(\frac{\partial T}{\partial t} + (\mathbf{u} \cdot \nabla) T \right) = -(\nabla \cdot \mathbf{q}) + \boldsymbol{\tau} : \mathbf{S} - \frac{T}{\rho} \frac{\partial \rho}{\partial T} \bigg|_p \left(\frac{\partial p}{\partial t} + (\mathbf{u} \cdot \nabla) p \right) + Q \quad (2.10)$$

In our case, since the fluid is Newtonian and incompressible, these equations can be simplified into

$$\begin{aligned} \rho \frac{\partial \mathbf{u}}{\partial t} + \rho (\mathbf{u} \cdot \nabla) \mathbf{u} &= \nabla \cdot [-p \mathbf{I} + \mu (\nabla \mathbf{u} + (\nabla \mathbf{u})^T)] + \mathbf{F} \\ \rho \nabla \cdot \mathbf{u} &= 0 \end{aligned} \quad (2.10)$$

Though the derivation of above equations is beyond the scope of our research, it must be mentioned that these equations are used in the computational fluid dynamics programs (CFDs) we are using in the next section to calculate the numerical solutions to our aerodynamic models.

Model

3.1 Theoretical Model

When the water jet of radius a hits the disk with velocity v_0 , its instantaneous volume flow rate is:

$$Q_0 = \pi a^2 v_0 \quad (3.1)$$

By the principle of continuity of fluid flow,

$$Q_0 = Q_1 + Q_3 \quad (3.2)$$

Where Q_1 is the volume flow rate of water along the axial direction, and Q_3 is the volume flow rate of water through the hole.

Hence, we can get

$$Q_1 = \pi a^2 v_0 - \pi b^2 v_3 \quad (3.3)$$

From our experimental results and our CFD results, it is reasonable for us to assume that $v_3 \approx v_0$. This assumption will be proved in section 3.2.2. Hence we can roughly calculate the axial volume flow rate

$$Q_1 = \pi(a^2 - b^2)v_0 \quad (3.4)$$

As the jet hits the surface of the disk, it applies a downthrust on the disk. The magnitude of this force can be calculated through the fluid momentum equation as well. The downthrust F_d can be calculated as the total rate of loss of fluid momentum in the vertical direction. Since the water current spreading out axially has lost all their vertical momentum,

$$F_d = \rho\pi(a^2 - b^2)v_0^2 \quad (3.5)$$

Assuming the friction between the water current and the disk surface is negligible, by the principle of continuity of fluid, where h is the depth of the water along the axial direction at distance r to the centre of the disk,

$$Q_1 = 2\pi r h v_1 \quad (3.6)$$

Birkhoff and Zarantonello(1957)[6] proved that when h is small and r is large as compared to the radius of the jet a , $v_1 \approx v_0$. Hence by equation (3.4) and (3.5), we are able to get

$$h = \frac{a^2 - b^2}{2r} \quad (3.7)$$

Hence by integration, the total volume of water spreading horizontally on the disk

$$V_{water} = \int_b^R 2\pi r h \, dr = \int_b^R 2\pi r \left(\frac{a^2 - b^2}{2r} \right) dr = \pi(R - b)(a^2 - b^2) \quad (3.8)$$

Total Weight of the water on the disk

$$W_{water} = \rho V_{water} g = \rho \pi g (R - b)(a^2 - b^2) \quad (3.9)$$

Combining with equation(3.5), the total normal force water exerts on the disk surface is

$$N_{total} = W_{water} + F_d = \rho \pi g (R - b)(a^2 - b^2) + \rho \pi (a^2 - b^2) v_0^2 \quad (3.10)$$

, which is downwards in direction.

By Equation(3.7), the depth of water at the start of the hydraulic jump

$$h_1 = \frac{a^2 - b^2}{2R} \quad (3.11)$$

Substituting Equation(3.9) into equation(2.7), we can get the depth of water after the hydraulic jump is

$$h_2 = \frac{a^2 - b^2}{4R} \left(-1 + \sqrt{1 + \frac{16Rv_0^2}{g(a^2 - b^2)}} \right) \quad (3.12)$$

Hence the Upthrust due to hydraulic jump can be calculated as

$$U_{total} = \rho \pi g h_2 (R^2 - b^2) = \frac{\rho \pi g (a^2 - b^2)(R^2 - b^2)}{4R} \left(-1 + \sqrt{1 + \frac{16Rv_0^2}{g(a^2 - b^2)}} \right) \quad (3.13)$$

Since the weight of the disk is

$$W_{disk} = m_{disk} g \quad (3.14)$$

Hence balancing the total force on the disk,

$$W_{disk} + N_{total} = U_{total} \quad (3.15)$$

The maximum mass of the disk that can be supported by the hydraulic jump is given by

$$m_{disk} = \frac{\rho \pi (a^2 - b^2)(R^2 - b^2)}{4R} \left(-1 + \sqrt{1 + \frac{16Rv_0^2}{g(a^2 - b^2)}} \right) - \rho \pi (R - b)(a^2 - b^2) - \frac{\rho \pi}{g} (a^2 - b^2) v_0^2 \quad (3.16)$$

3.2 CFDs Model

3.2.1 Model Setup

COMSOL, a Multiphysics simulation platform, was used to develop our computational model. It uses Finite Element Method (FEM) to break our model into small elements. After that, iterations are carried out repeatedly to find solutions for our K - ϵ Turbulent Model.

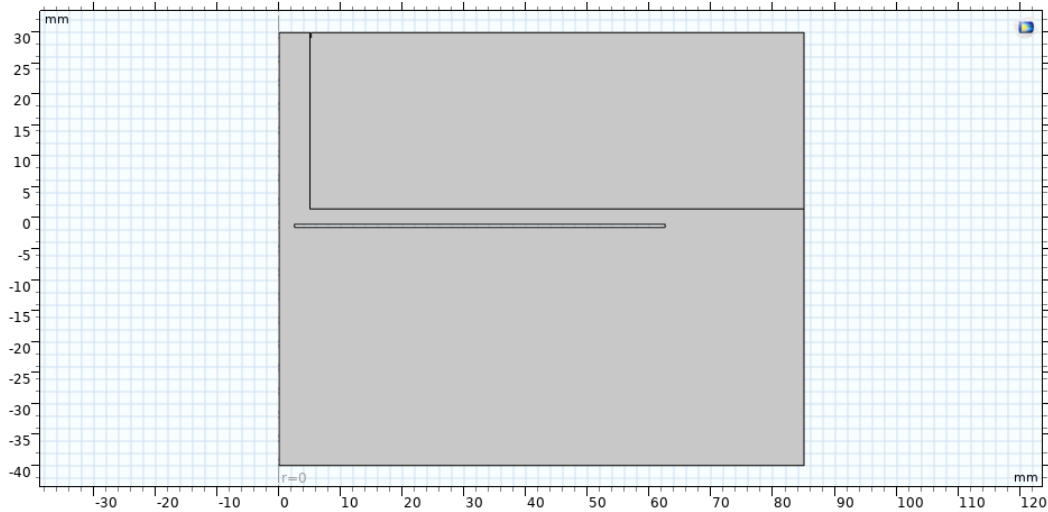


Figure 3.1: Model Setup

Assuming the water flow is axially symmetrical in our experiments, I can simplify our model into a two-dimensional flow along the axial direction. A model like Figure 3.1 is therefore set up to simulate the flow situation. The horizontal axis is the distance away from the centre of the disk along the axial direction, while the vertical axis. A rectangle Two-Phase Flow Field considering both the water flow and the resultant air flow is set up. The width of the flow field is set to be 85mm, which is the radius of the cylindrical water tank I am going to use later in my experiment. The height of the flow field is set to be as large as possible so that we can obtain correct height of hydraulic jump under any condition. Also, a large water depth can reduce the amount of water current being reflected by the base of the tank onto the lower surface of the disk, which may cause the upthrust by the disk calculated by water to be inaccurate.

The resultant flow pattern when the inlet velocity $v_{in} = 1.11 \text{ m s}^{-1}$ (which corresponds to $v_0 = 1.35 \text{ m s}^{-1}$), jet radius = 0.005 m, hole radius = 0.0025 m, disk radius = 0.0625 m and temperature =

303.15 K is shown below.

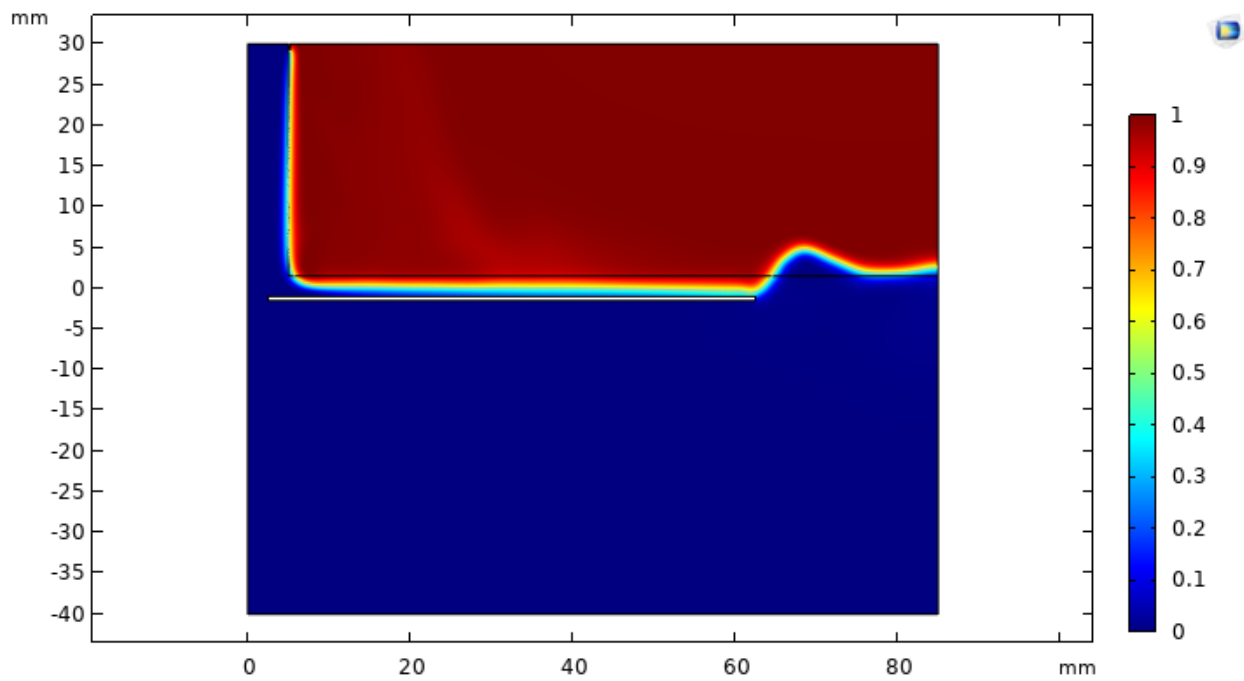


Figure 3.2: 2D flow pattern

From the graph, we can calculate the hydraulic jump produced in this case to be 0.0075 m. The generated 3D flow pattern is shown below:

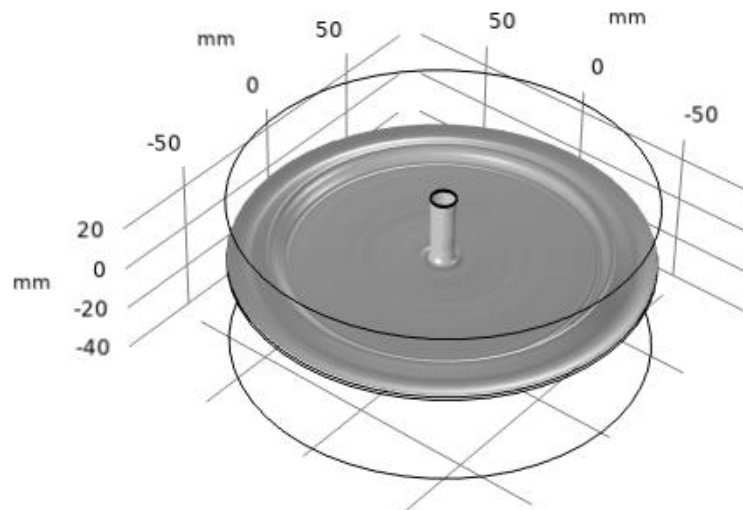


Figure 3.3: 3D flow pattern

Two boundary probes to measure the total force acting on the surface were placed at the upper surface and the lower surface of the disk respectively(Fig 3.4).

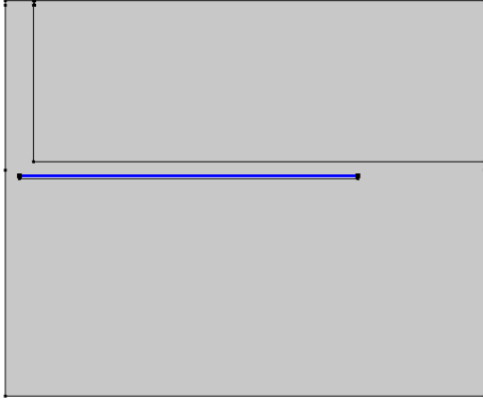


Figure 3.4a: Upper surface boundary probe

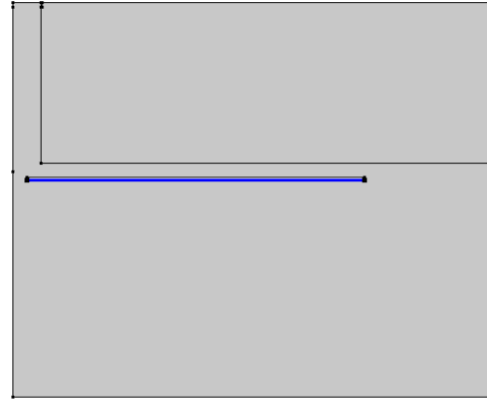


Figure 3.4b: Lower surface boundary probe

Hence, we can get the net force acting on the disk to be the value returned by the upper surface boundary probe subtracted by the value returned by the lower surface boundary probe subtracted. A negative value of this net force will denote a net upthrust on the disk. Record this value as F_t , then the maximum mass of the disk the hydraulic jump can support is calculated by

$$m_{disk} = \frac{F_t}{g} \quad (3.17)$$

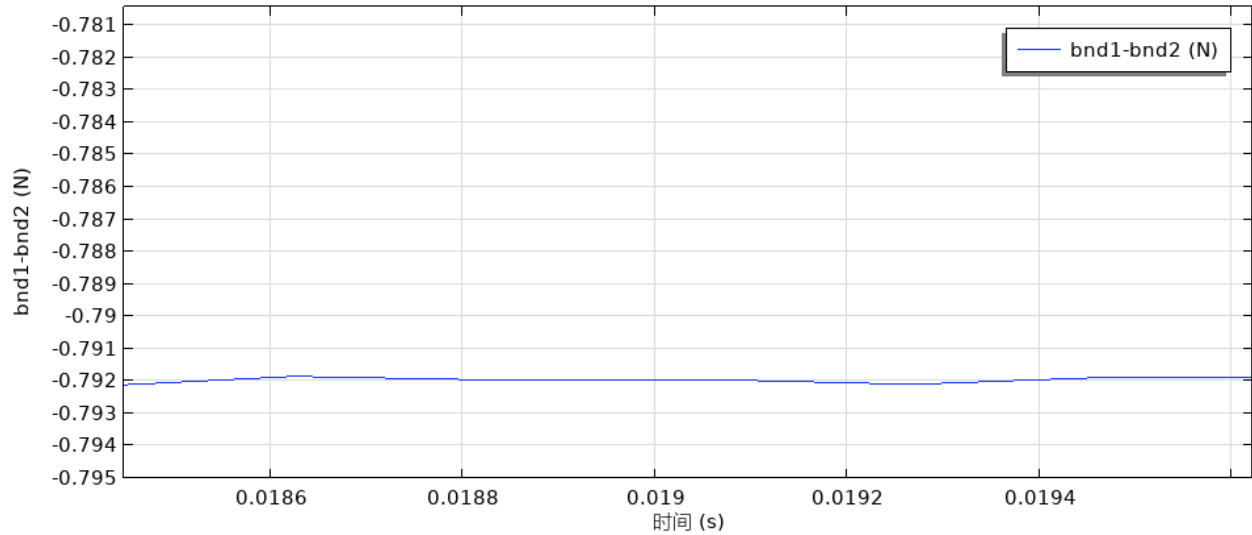


Figure 3.5: Net force on the disk

Hence from the data we can calculate the average net force on the disk to be 0.79 N , which can support disk of maximum mass 0.081 kg .

3.2.2 Verifying Assumptions for Theoretical Model

The CFDs model can also be used to verify our key assumption for the theoretical model, which is $v_0 \approx v_1 \approx v_3$.

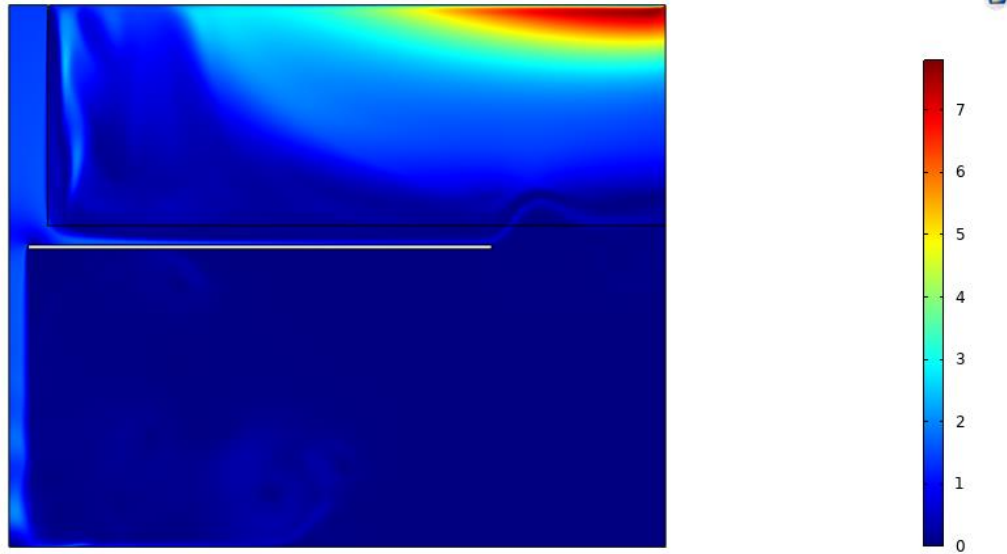


Figure 3.6: 2D velocity pattern around the disk

From the velocity graph we can roughly observe that $v_0 \approx v_1 \approx v_3$ holds true.

This assumption can be proven when we add two domain point probes at 1.5 mm distance near the disk's hole that measures the local velocity(Fig 3.7), and calculate the ratio of the value they returns

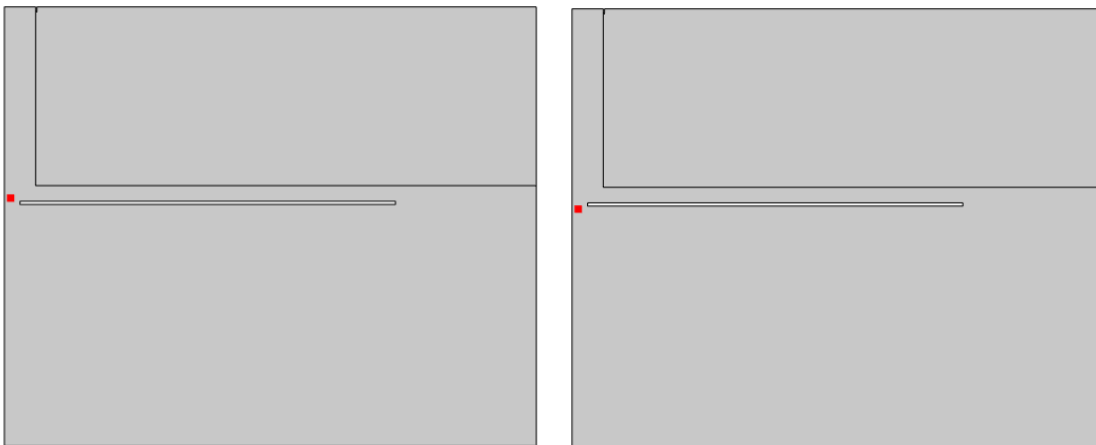


Figure 3.7: Domain point probes

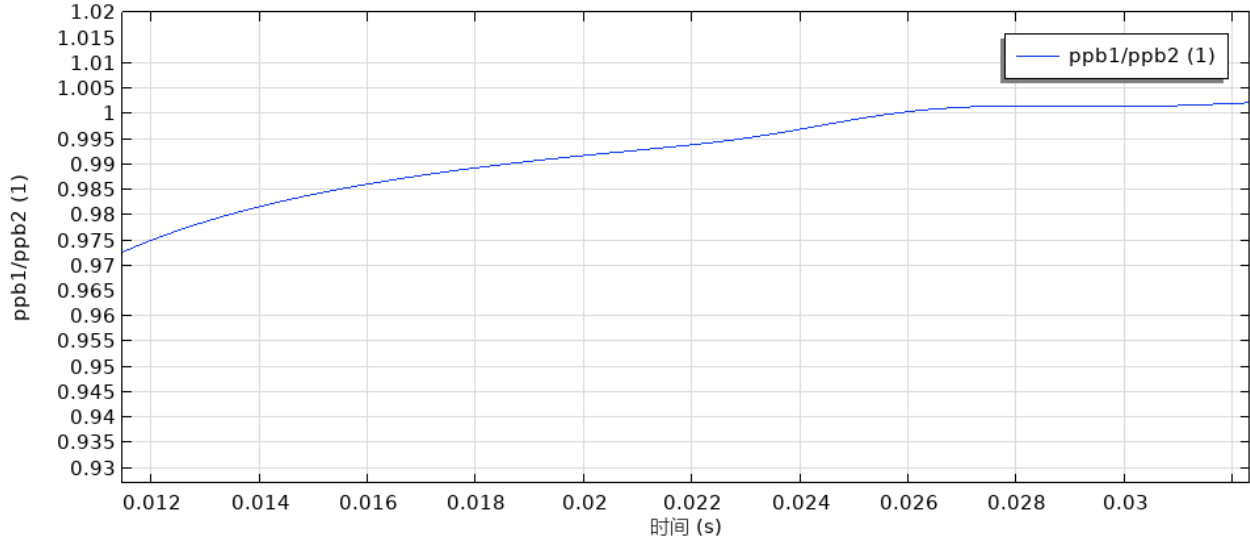


Figure 3.8: Ratio of the value returned by the point probes

Since the ratio of the values returned by the two probes is nearly one, it is reasonable for us to assume $v_0 \approx v_1 \approx v_3$ in our theoretical model

3.2.3 Model Assumption and Limitation

The model is developed assuming the flow is axially symmetrical. However, in reality there are factors such as surface waves that may not affect the disk in an axially symmetric manner. The model is therefore unable to predict the effect of such factors on the disk.

Experiment

The experiments were conducted in an outdoor area outside the physics laboratory of the Science Research Centre of Hwa Chong Institution. Since Singapore's daytime temperature is relatively consistent and approximately 30 degrees Celsius, the value of temperature in our CFD model and theoretical calculation is set to be 303.15K.

4.1 Experimental Setup



Figure 4.1 Experimental Setup

My experimental setup is shown in Fig. 4.1. A double water pump system has been established with holes at the side of the container. This helps to ensure that the water level at the tank is constant and always equal to the height of the hole, and the rate of flow of water jet can be kept constant for a longer time. The overflowing situation in the cylindrical tank minimises water perturbation, thereby reducing the error due to surface waves. The acrylic tubes and the water pumps were connected by 3D printed connecting tubes and sealed with tape to reduce water leak. There is also a 3D printed nozzle installed at the end of the acrylic tube to accurately change the diameter of the water jet. The height of the hydraulic jump is measured through a stainless-steel protractor ruler held vertical by the retort stand.



Figure 4.2: Various pumps and 3D printed nozzles used in my experiment

Various water pumps of different maximum flow rates and a large pump with adjustable flow rate are used to generate different jet velocity required in my experiment. I also used 3D printer to print nozzles of outlet radius 3mm to 4.5mm to change the radius of the water jet produced.



Figure 4.3: Various disks of different radius and hole radius used in my experiment

Various disks were laser cut from a uniform aluminium sheet of 0.5 mm thickness. These disks with radius ranging from 27.5 mm to 77.5 mm , hole radius ranging from 1 mm to 7 mm , were used in my experiment.

4.2 Data Collection Processes



Figure 4.3: Various washers and weights attached to the Blu Tack in my experiment

Since due to constraint in apparatus and the turbulent nature of the hydraulic jump, it is hard to measure the height of hydraulic jump very accurately. Hence, a more accurate approach to verify the model is introduced——Measuring the maximum mass of the disk the hydraulic jump can support.

However, it is complicated and not economically viable to use disks of so many different densities. I adopted an alternative measure by sticking the Blu Tack to the disk to form a complete loop around the hole, and attaching washers and weights of different masses to it. The washers and weights were added in an axially symmetrical manner to ensure the balance of the disk. Since the masses of individual washer can be as small as 0.3 g. We are able to achieve high precision in controlling the mass of the disk.

In my following experiments, the maximum mass of the disk that can be supported by the hydraulic jump was determined accurately through repeated experiments adjusting the mass of the disk by adding, substituting or removing washers and weights. The maximum mass that can allow the disk to stay afloat is then recorded.

To make sure the disk does not get carried away by waves and deviate from the centre of the water jet, a very small dent was made at the centre of the disk. When the disk undergoes a small displacement due to water perturbation, it would produce an additional torque to restore the disk back to its original position exactly under the water jet. This helps to stabilise the effect, allowing us to obtain more accurate data.

4.3 Experimental Data Analysis

In the following data analysis, the maximum mass of disk the hydraulic jump can support is plotted against the variable to be investigated. Two sets of data which are the CFD model data and experimental data will be plotted together with the theoretical curve for comparison and analysis.

4.3.1 Jet Velocity

When investigating the factor jet velocity, the other factors are kept constant to: $a = 0.005 \text{ m}$, $b = 0.0025 \text{ m}$, $R = 0.0625 \text{ m}$, while the jet velocity changes from 1.55 m s^{-1} to 6.70 m s^{-1} .

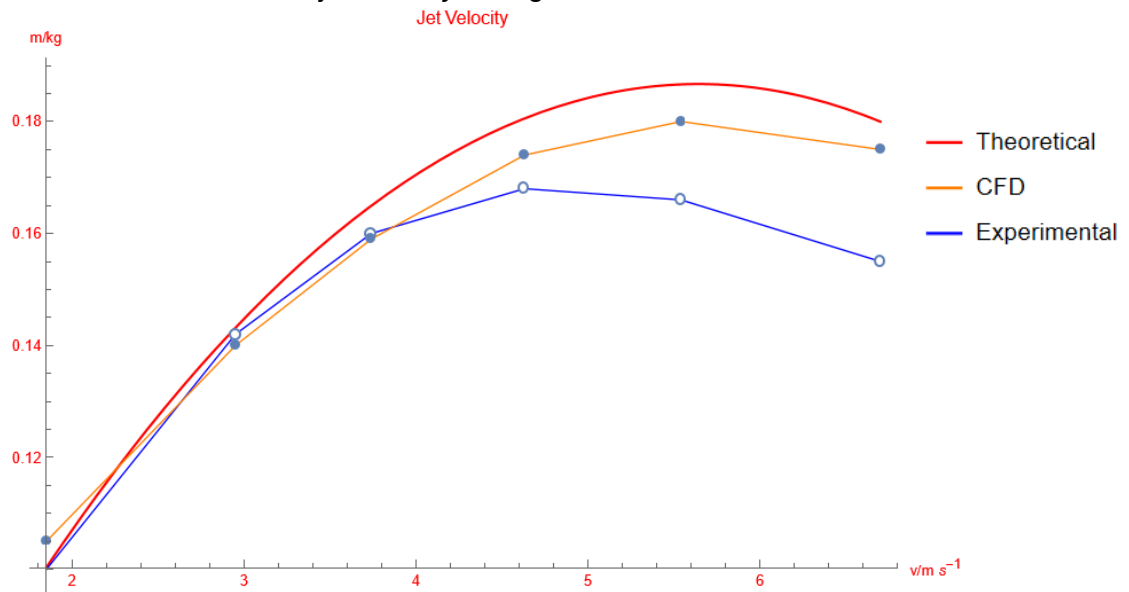


Figure 4.4: Jet velocity experimental data

As shown in the graph, the maximum mass of disk the hydraulic jump initially has a positive relationship with the jet velocity. However, as the jet velocity continue to rise, the maximum mass of the disk the hydraulic jump can support will eventually fall.

This could be because as the jet velocity continue to rise, the downthrust of the water jet applies to the disk is increasing at a faster rate than the upthrust provided by the hydraulic jump, resulting in a net decrease in the maximum mass of the disk the hydraulic jump can support.

The experimental data and CFD model data generally show good coherence with the theoretical data, especially when the jet velocity is small.

However, at large jet velocities, the maximum mass of disk observed by experiments and calculated by CFD tend to be smaller than the theoretical value. This could be because a larger jet velocity can create stronger surface waves, causing fluctuations in the upthrust provided by

the hydraulic jump. Since a temporary decrease of upthrust could lead to the disk sinking before the actual theoretical threshold reaches, the maximum mass of disk the hydraulic jump can support being lower than expected.

4.3.2 Disk Radius

When investigating the factor disk radius, the other factors are kept constant to: $a = 0.005 \text{ m}$, $b = 0.0025 \text{ m}$, $v_0 = 1.85 \text{ m s}^{-1}$, while the disk radius changes from 0.0275 m to 0.0775 m .

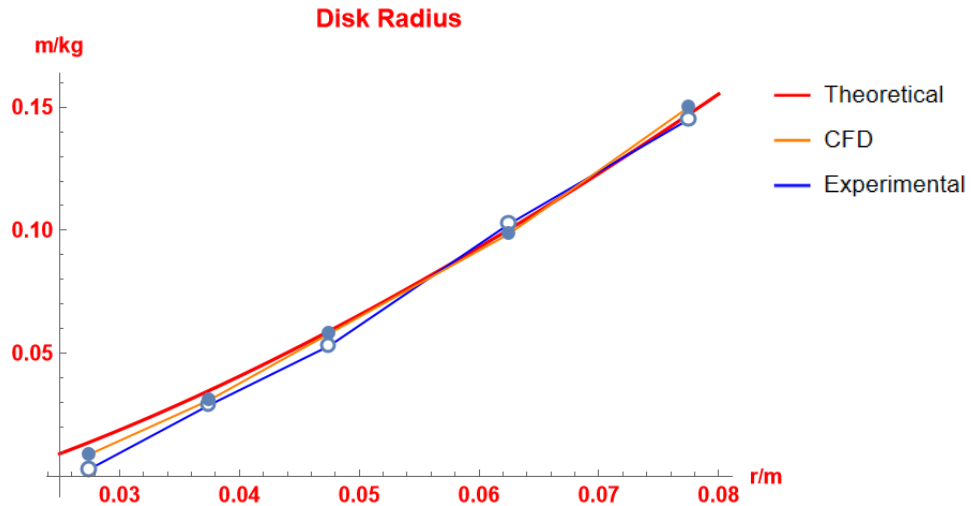


Figure 4.5: Disk radius experimental data

As shown in the graph, the maximum mass of disk the hydraulic jump has a positive relationship with disk radius.

This could be because as the disk radius continue to rise, the upthrust provided by the hydraulic jump on the disk will increase, resulting in an increase in the maximum mass of the disk the hydraulic jump can support.

The experimental data and CFD model data shows adequate coherence with the theoretical data, especially when the disk radius is large.

However, at small disk radius, the maximum mass of disk observed by experiments and calculated by CFD tend to be a lot smaller than the theoretical value. This could be because disk of smaller are more seriously affected by water perturbations. Also, based on my observation, the instantaneous initial downthrust on the disk may be too large for small disks and the small disks sink before the hydraulic jump could form. Hence, this leads to the maximum mass of disk the hydraulic jump can support being lower than expected.

4.3.3 Jet Radius

When investigating the factor jet radius, the other factors are kept constant to: $r = 0.0475 \text{ m}$, $b = 0.0025 \text{ m}$, $v_0 = 5.10 \text{ m s}^{-1}$, while the jet radius changes from 0.005 m to 0.003 m .

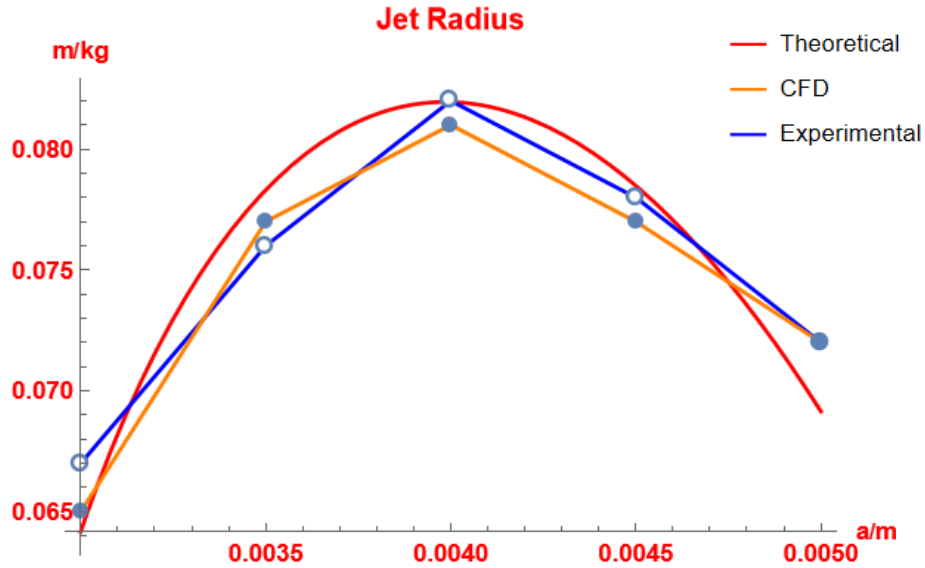


Figure 4.6: Jet radius experimental data

As shown in the graph, the maximum mass of disk the hydraulic jump initially has a positive relationship with the jet radius. However, as the jet radius continue to rise, the maximum mass of the disk the hydraulic jump can support will eventually fall.

Similarly, this could be because as the jet radius continue to rise, the downthrust of the water jet applies to the disk is increasing at a faster rate than the upthrust provided by the hydraulic jump, resulting in a net decrease in the maximum mass of the disk the hydraulic jump can support.

The experimental data and CFD model data generally show good coherence with the theoretical data. The error mainly comes from the strong water perturbation caused by high jet velocity used in the experiment.

4.3.4 Hole Radius

When investigating the factor hole radius, the other factors are kept constant to: $r = 0.0475 \text{ m}$, $a = 0.005 \text{ m}$, $v_0 = 4.20 \text{ m s}^{-1}$, while the hole radius changes from 0 m to 0.007 m .

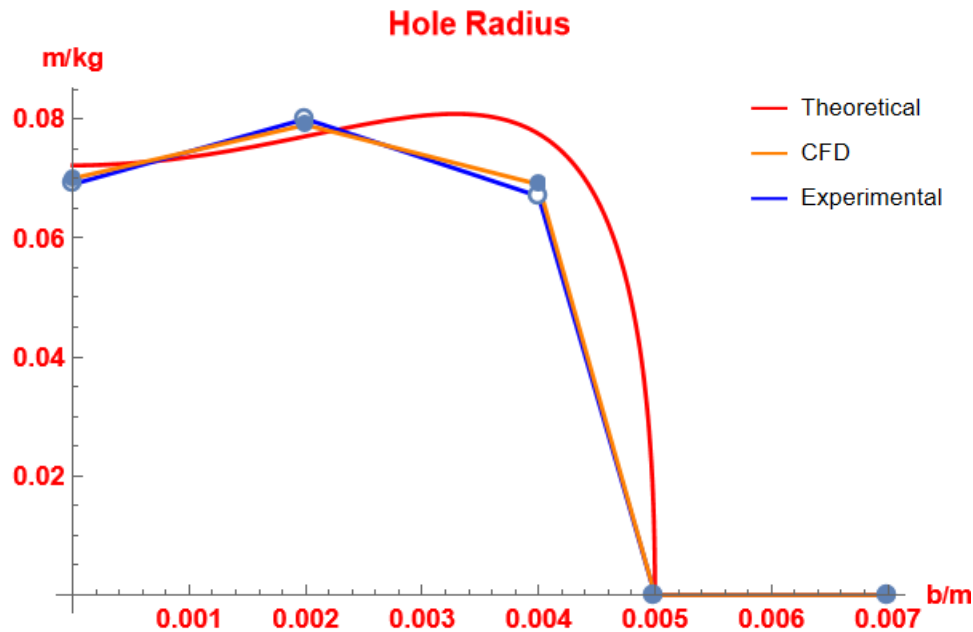


Figure 4.6: Hole radius experimental data

As shown in the graph, the maximum mass of disk the hydraulic jump initially increases slowly as the hole radius increases. However, as the hole radius continue to rise, the maximum mass of the disk the hydraulic jump can support will eventually fall rapidly. When the hole radius is larger than the jet radius, the disk will sink.

This could be because as the hole radius rises initially, the downthrust of the water jet applies to the disk is decreasing at a slightly faster rate than the upthrust provided by the hydraulic jump, resulting in a gradual increase in the maximum mass of the disk the hydraulic jump can support. However, as the hole radius approaches the magnitude of the jet radius, the upthrust is decreasing at a much faster rate than the downthrust, resulting in a rapid decrease in the maximum mass of the disk the hydraulic jump can support.

The experimental data and CFD model data generally show good coherence with the theoretical data. Due to time and apparatus constraint, only five data points are obtained in this experiment. More data points could be collected to verify the theory better

4.4 Limitations of Model

The model cannot address factors such as water perturbation and the resultant oscillation of the disk into account. The surface waves could result in a temporary decrease in upthrust experienced by the disk, causing the disk to sink even if their mass did not reach the threshold value. Oscillation may cause the disk to tilt and sink due to unevenly distributed forces. Hence these factors may lead to inaccuracy of our theoretical model.

4.5 Error Analysis

4.5.1 Inaccuracy in Measurement

Due to budget constraints, in this experiment I could only use a stainless-steel protractor ruler to measure the height of the hydraulic jump. The actual flow rate of each pump was determined using repeated measurements with a large 500ML measuring cylinder and stopwatch. This could lead to inaccuracy in my measurement.

4.5.2 Air Entrainment in the Jet

As the water jet leaves the nozzle/end of the acrylic tube, it continues to accelerate downwards in the air. This will result in air blending in with the jet. When it impacts the disk surface, the pressure due to the air inside the jet being compressed may produce an additional downward force on the disk, leading to the maximum mass of disk the hydraulic jump can support measured being less than expected.

4.6 Possible Improvements

4.6.1 Using More Accurate Apparatus



Figure 4.7a: Point gauge



Figure 4.7b: Digital flow meter

A point gauge (Fig. 4.7a), which is a device used to measure water levels, could be installed to measure the height of the hydraulic jump. The adjustable, pointed metallic rod on it can be lowered until the moment it pierces the water surface[10]. A digital flow meter (Fig. 4.7b) could also be installed at the end of the acrylic tube to accurately measure the jet volume flow rate. Hence these two apparatuses can help to determine the height of the hydraulic jump and the jet velocity more accurately to better verify our theoretical model.

4.6.2 Improving the Experimental Setup

The experimental setup can be improved by further reducing the height of the end of the acrylic tube and reducing the length of acrylic tube in which the water flow downwards. This will allow the water to experience less acceleration due to gravity, leading to less air entrainment in the jet impacting the disk surface.

Conclusion

In conclusion, the maximum mass of the disk the hydraulic jump can support is in positive relationship with the disk radius, but only in positive relationship with jet velocity, jet radius and hole radius initially. As the jet velocity, jet radius and hole radius increases pass a certain threshold, the maximum mass of the disk the hydraulic jump can support will decrease.

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